

NX-Command 1

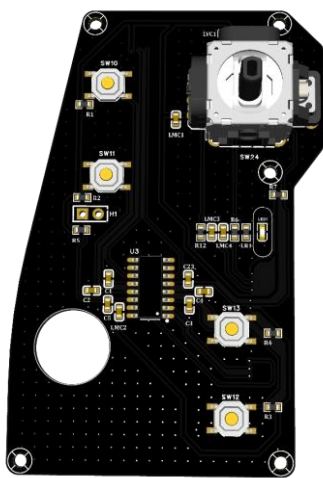
Dual-Mode UAV Controller

Hall Effect Joystick • Split-PCB Design • SBUS / USB HID Dual Mode | **Made in India**

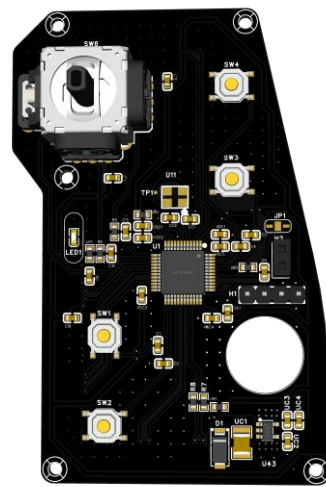
4 Analog Axes	16 Switches (Gamepad)	USB HID Gamepad	Split-PCB 2× Board
Hall Effect Drift-Free	12 RC Switches (RC Mode)	SBUS Serial RC Output	UART / ETH Radio Ready

Product Images

NX-Controller (Ground Side)

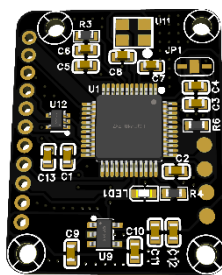


Left Hand Side (LHS) Board



Right Hand Side (RHS) Board

NX-SBus (Air Side)



Sbus-Top



Sbus-Bottom

General Description

The NX-Controller is a dual-mode gamepad controller module purpose-built for Unmanned Aerial Vehicle (UAV) ground control applications. Employing hall-effect sensing technology on both axes, the NX-Controller delivers smooth, precise, drift-free analog input across the full operating life of the product. The controller is physically partitioned into two matched PCBs - Left Hand Side (LHS) and Right Hand Side (RHS) - enabling flexible mechanical integration into custom GCS enclosures, display-based controllers, and handheld consoles.

Two distinct operating personalities are firmware-selectable:

- **Gamepad Mode** - The device enumerates as a standard USB HID gamepad. It is immediately compatible with Mission Planner, QGroundControl, and any MAVLink-override-capable GCS, requiring no additional drivers on Windows, macOS, or Linux.
- **RC Mode** - Joystick axes and button inputs are converted in real-time to the industry-standard SBUS serial format and output over UART. When paired with a ground-station radio transmitter and an airborne SBUS decoder, the controller provides a complete RC-link solution without a dedicated RC transmitter.

Key Features

✓ Hall Effect - contactless, drift-free, lifetime-rated
✓ 4 × Analog (Left X/Y, Right X/Y) — 12-bit resolution
✓ Split 2-PCB design for easy display-controller integration
✓ 16 × programmable buttons
✓ 12 RC Switch channels: 4 for Flight Mode select, 2 for ARM/DISARM
✓ USB HID Gamepad - Compatible with all major GCS software
✓ SBUS (standard 100 kbaud inverted UART)
✓ UART SBUS output for radio integration (UART / Ethernet radio)
✓ On-board LED (LHS & RHS) for power indication
✓ Any UART or Ethernet-based radio module
✓ 5 V DC (USB) / 4.8 V–5.2 V (external)
✓ Made in India

Operating Modes

Gamepad / USB HID Mode

- Controller enumerates as a standard USB HID Gamepad on any host computer.
- All 4 analog axes are reported at full 12-bit resolution.
- 16 buttons available:
- Compatible with Mission Planner, QGroundControl, and any GCS that supports MAVLink RC Override.
- No driver installation required (Windows / macOS / Linux plug-and-play).
- Powered directly from the USB host at 5 V; no external supply needed.

RC / SBUS Mode

- 4 × Flight Mode select buttons (mapped to SBUS channel 5).
- 2 × ARM / DISARM dedicated buttons (mapped to SBUS channel 6).
- 12 RC Switches total in this mode (4 flight-mode combined, 2 ARM/DISARM, 6 user-assignable).
- SBUS output on UART_TX: 100 kbaud, 8E2, inverted — standard Futaba SBUS specification.
- 16 SBUS channels are always transmitted.
- An external SBUS-capable board is required in the airborne radio to decode SBUS and forward to the FC RCIN port.

System Integration Diagram

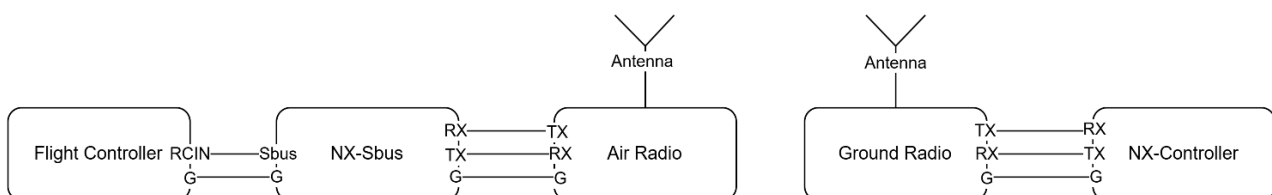
RC Mode - Ground to Air Link:

NX-CONTROLLER	Ground Radio	→	Air Radio	NX-SBUS	Flight Controller
UART / Ethernet	UART / Ethernet	→	UART / Ethernet	UART / Ethernet	RCIN/SBUSIN

Gamepad Mode - USB HID Link:

NX-CONTROLLER	→	GCS Computer	→	FC via MAVLink Override
USB HID Gamepad	→	Mission Planner / QGC	→	Telemetry

Typical Diagram



Electrical Specifications

Parameter	Min	Typ	Max	Unit
Supply voltage (USB)	4.8	5.0	5.2	V
Supply current (idle)	—	80	150	mA
Supply current (active)	—	120	200	mA
Joystick axis resolution	—	12	—	bits
UART_TX voltage level	—	3.3	—	V
SBUS baud rate	—	100	—	kbaud
Operating temperature	−20	25	+70	°C
USB D+ / D− signal level	—	3.3	—	V

Connector Pin-Out

The NX-Controller exposes two connector groups — a Power/USB group and a Serial group — accessible via the board-edge connectors.

Power & USB Connector

Pin	Signal	Type	Description
1	5V	Power In	5 V supply input. Powers the controller when not using USB bus power
2	USB+	USB D+	USB data positive (D+)
3	USB−	USB D−	USB data negative (D−)
4	GND	Ground	Common ground reference

Serial / RC Connector

Pin	Signal	Type	Description
1	UART_RX	Digital In	UART receive (3.3 V logic level).
2	UART_TX	Digital Out	UART transmit (3.3 V logic level).
3	GND	Ground	Serial ground reference.
4	GND	Ground	Additional ground.

Typical Applications

- Fixed-wing, multirotor, and VTOL UAV ground control stations
- Custom GCS consoles with integrated display (USB HID Joystick Mode)
- Long-range UAV operations using UART / Ethernet radio links (RC Mode)
- Rover and ground robot teleoperation
- Industrial inspection drone remote control
- Development and prototyping of UAV control systems using MAVLink

Ordering Information

Part Number	Product Description
NX02-01	Controller Unit
NX02-02	Sbus Unit

Custom boards and bulk pricing available on request.

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